

Parallel and Distributed Computing

Study Guide (Lecture 2 Focus)

Part 1: Past Papers

QUIZ 1 SPRING 2025 • QUESTION # 1

Q: Write two major differences between shared-memory and message-passing based parallel and distributed systems.

A:

- **Address Space:** Processors in a shared-memory system share the same physical address space (often kept coherent by hardware when multiple processors read/write to the same location). In contrast, processors in a message-passing system have their own private memories and disjoint address spaces.
- **Data Interaction:** In shared memory, communication occurs implicitly by modifying shared variables, whereas in message passing, explicit sending and receiving of messages is required to transfer data between processors.

QUIZ 1 SPRING 2025 • QUESTION # 2

Q: Name a hardware for each of the following four, which can be classified as: (a) SISD, (b) SIMD, (c) MIMD, (d) MISD.

A:

- **SISD:** Single-core Pentium Pro processor.
- **SIMD:** Vector processors, or SSE / AVX execution units in Intel processors.
- **MIMD:** Multi-core processors (e.g., modern multi-core CPUs).
- **MISD:** No standard commercial hardware exists; Systolic arrays (e.g., in Google's TPU) or flight control hardware in NASA's shuttles can be considered examples.

QUIZ 01 FALL 2022 • QUESTION # 1

Q: We would call a set of _____ computers that may run different OSes as _____.

- a. co-located; a static network
- b. dispersed; network of workstations
- c. co-located; network of workstations
- d. dispersed; cluster

A: b. dispersed; network of workstations

Q: Most modern day parallel architectures would fall under which category of Flynn's taxonomy?

- a. SISD
- b. MISD
- c. SIMD
- d. MIMD

A: d. MIMD

Q: How can we classify the type of computer architecture on a typical, consumer-grade multi-core laptop computer released in the year 2024 out of the following?

- A) SISD
- B) SIMD
- C) MIMD
- D) MISD

A: (A), (B), and (C). If executing a single sequential thread, it operates as SISD; if executing across all cores concurrently, it acts as MIMD; and modern CPUs include vector/SIMD instruction extensions (e.g., AVX), hence SIMD.

Q: Interpretation of Moore's law asserts that:

- A) Computing power will eventually level off
- B) Computing power doubles every 18 months
- C) Transistors decrease in size 50% every 2 years
- D) Data storage costs decrease by 50% every 18 months

A: B) Computing power doubles every 18 months.

Q: Different computers possess the same version of operating system when connected in the form of:

- A) Cluster
- B) Cloud
- C) Network of workstations
- D) None of the given options

A: A) Cluster.

MIDTERM I SPRING 2022 • QUESTION # 1

Q: Single Program with Multiple Data (SPMD) is a close variant of _____ in Flynn taxonomy:

- a. SIMD**
- b. MIMD**
- c. MISD**
- d. SISD**

A: b. MIMD.

MIDTERM I SPRING 2022 • QUESTION # 2

Q: Choose correct statement(s) regarding Network of Workstations:

- a. The workstations in such a network may have different operating systems**
- b. The nodes within a network will normally be co-located (located together)**
- c. Users normally have the power to login to individual workstations**
- d. Both (a) and (c)**

A: d. Both (a) and (c).

MIDTERM I SPRING 2022 • QUESTION # 3

Q: In distributed Systems, processors can share logical address space. (True/False)

A: False. Processors in distributed systems have disjoint logical and physical address spaces.

MIDTERM I SPRING 2023 • QUESTION # 1

Q: GPU is an example of:

- a. Shared Memory SIMD**
- b. Distributed Memory SIMD**
- c. Shared Memory MIMD**
- d. Distributed Memory MIMD**

A: a. Shared Memory SIMD.

MIDTERM I SPRING 2023 • QUESTION # 2

Q: We can increase performance of a uniprocessor by:

- a. Increasing clock frequency**
- b. Increasing instructions per cycle**
- c. Multithreading**
- d. Any one from above can be used**

A: d. Any one from above can be used.

MIDTERM II SPRING 2023 • QUESTION # 1

Q: With distributed systems, we will have:

- a. Processors with a shared logical address space
- b. Processors with a disjoint logical address space
- c. Processors that interact with each other by message passing
- d. Both (b) and (c)

A: d. Both (b) and (c).

MIDTERM II SPRING 2023 • QUESTIONS # 2-5 & MID II RETAKE SPRING 2023 • QUESTION # 1

Q: Operating Systems for Distributed and Parallel Computing:

- Which OS hides distribution completely and presents a single system image?
- Which OS is suited as tightly coupled OS for multiprocessors/homogeneous multi-computers?
- Which OS is suited as loosely coupled OS for heterogeneous multi-computers?
- eScience applications are suited for which type of computing?
- True/False: Distributed Operating Systems are better suited for heterogeneous multi-computers.

A:

- Hides distribution: **Middleware**.
- Tightly coupled / Homogeneous: **Distributed OS**.
- Loosely coupled / Heterogeneous: **Network OS**.
- eScience: **Grid Computing**.
- False (Distributed OS is for homogeneous; Network OS/Middleware is for heterogeneous).

ASSIGNMENT 2 SPRING 2023 • QUESTION # 1

Q: Provide a comparison of Network Operating System, Distributed Operating System, and Middleware. Briefly explain the goals of a middleware platform.

A:

- **Network Operating System (NOS):** Loosely coupled OS designed for heterogeneous multi-computers where individual nodes run independent operating systems.
- **Distributed Operating System (DOS):** Tightly coupled OS designed for multiprocessors and homogeneous multi-computers, managing system hardware uniformly.
- **Middleware:** A software layer placed above local operating systems that hides physical distribution and heterogeneous details.
- **Goals of Middleware:**
 - **Resource Sharing:** Enables accessing and sharing resources seamlessly across a distributed network.
 - **Transparency:** Provides the illusion of viewing the entire distributed network as if it were a single integrated computer system.

QUIZ 05 SPRING 2022 • QUESTION # 1

Q: Draw an image of two systems, one where you would prefer using OpenMP for parallel programming, and the other where you would prefer using MPI.

A:

- **For OpenMP:** Illustrate a shared memory architecture where all processors connect directly to a single shared physical memory unit.
- **For MPI:** Illustrate a distributed memory architecture where independent computing nodes (processor + local memory) connect via a network interconnection.

SESSIONAL-I EXAM SPRING 2025 / FEB 24TH 2025 • QUESTION # 1

Q: You are designing a cloud-based image processing service where each uploaded image is independently transformed (e.g., resized and enhanced). With thousands of concurrent uploads, which Flynn's Taxonomy architecture best supports parallelism and scalability? (Just name it.) Provide rationale for your answer in a line.

A: MIMD (or SPMD / SIMD):

- **MIMD:** Multiple independent processors execute operations on independent image data streams concurrently.
- **SPMD / SIMD:** All processing units execute the same image transformation algorithm across distinct data inputs.

FINAL EXAM SPRING 2025 • QUESTION # 1

Q: A processor executes a program that consists of 2 billion instructions. The system reports the following performance metrics during execution:

- **Wall-clock time (total elapsed time): 5 seconds**
- **CPU time (user + system time): 3.8 seconds**
- **System overhead time (I/O waits, context switches, etc.): 1.2 seconds**

The processor's clock rate is 2.5 GHz. It is known that 25% of the instructions are floating-point operations.

a. Calculate the MIPS rating of the processor. Clearly justify which time value you used in your calculation and why.

b. Using the appropriate timing, calculate the number of FLOPS achieved.

A:

a. $MIPS = (Instruction\ Count) / (Execution\ Time \times 10^6) = (2 \times 10^9) / (3.8 \times 10^6) = 526.31\ MIPS.$

Justification: CPU time (3.8 seconds) is used because MIPS measures the instruction execution rate of the processor core itself, excluding external system overheads like I/O waits and context switching.

b. Floating Point Operations = $25\% \times 2 \times 10^9 = 5 \times 10^8\ FLOPS.$

$FLOPS = (5 \times 10^8) / 3.8\ s \approx 1.315 \times 10^8\ FLOPS\ (or\ 131.5\ MFLOPS).$

Part 2: Book Questions (Introduction to Parallel Computing)

CHAPTER # 1 • QUESTION # 1.1

Q: Go to the Top 500 Supercomputers site (<http://www.top500.org/>) and list the five most powerful supercomputers along with their FLOPS rating.

A: This is an empirical research assignment requiring students to access www.top500.org and document the top 5 supercomputers alongside their Rmax / Rpeak FLOPS metrics.

CHAPTER # 1 • QUESTION # 1.2

Q: List three major problems requiring the use of supercomputing in the following domains: Structural Mechanics, Computational Biology, Commercial Applications.

A:

- **Structural Mechanics:** Crashworthiness simulation of automobiles, seismic vulnerability analysis of large dams/bridges, and stress analysis of aerospace structures.
- **Computational Biology:** Molecular dynamics modeling of protein folding, genomic sequence alignment, and rational drug design simulations.
- **Commercial Applications:** High-frequency automated stock trading analysis, credit card fraud detection in real-time banking, and web-scale search engine indexing / data mining.

CHAPTER # 1 • QUESTIONS # 1.3 & 1.4

Q: Collect statistics on the number of components in state of the art integrated circuits over the years. Plot the number of components as a function of time and compare the growth rate to that dictated by Moore's law. Repeat the experiment for the peak FLOPS rate of processors.

A:

- **1.3 (Transistors):** This exercise involves collecting historical IC transistor count metrics (from early chips like Intel 4004 to modern multi-core processors) and plotting them on a semi-log scale to verify the empirical exponential trend doubling component density every 18–24 months.
- **1.4 (FLOPS):** This exercise compares raw processing speed (FLOPS) growth against transistor growth. While single-core clock frequencies stalled due to power and thermal density walls, total multi-core peak FLOPS have sustained growth matching Moore's law via explicit hardware parallelism.

Q: What are the major differences between message-passing and shared-address-space computers? Also outline the advantages and disadvantages of the two.

A:

- **Shared-Address-Space Computers:**

- *Differences:* Processors communicate via a common globally accessible address space.
- *Advantages:* Simple programming model; data sharing is implicit.
- *Disadvantages:* Hardware cache-coherence mechanisms are complex and difficult to scale to large core counts; requires explicit synchronization (locks/semaphores).

- **Message-Passing Computers:**

- *Differences:* Each node consists of an independent processor and exclusive private memory; processors communicate strictly via explicit messages.
- *Advantages:* Highly scalable; simpler hardware without global cache coherence requirements.
- *Disadvantages:* Harder to program; higher message startup latencies (t_s).

Q: Of the four PRAM models (EREW, CREW, ERCW, and CRCW), which model is the most powerful? Why?

A: The CRCW (Concurrent-Read Concurrent-Write) PRAM model is the most powerful. It allows multiple processors to simultaneously read from and write to the exact same memory location in a single clock cycle, enabling parallel algorithms to achieve the minimum execution time among all PRAM variants.